

# POSSIBLE MASS DISTRIBUTIONS IN THE NEBULAE OF OTHER SOLAR SYSTEMS

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**Abstract.** The supernova shell fragmentation model of solar system formation – previously shown to be successful in describing the mass distribution of our solar system – is used to calculate the mass distributions of other solar nebulae.

## 1. Introduction

The present model of solar system formation is constructed on the concept of supernova shell fragmentation: each fragment is ejected outward from the explosion site, and each subsequently cools, sweeps up additional mass, and collapses to form an individual solar system (Brown, 1970, 1971a; Brown and Gritz, 1987). Arnett (1985) has possibly found the mechanism of envelope ejection from type II supernovae.

In Brown and Gritz (1987) we found that the key equation (arising from conservation of mass and angular momentum per unit mass) to map the fragment mass to a solar nebula with a dominant central mass,  $M_c$ , is of the form

$$r_2^{1/2} L_0^{3/2} = (1 - a) D_x^2 - a D_y^2, \quad (1)$$

where

$$L_0 \equiv \frac{(GM_c)^{1/3}}{(v_0/x_0)^{2/3}}. \quad (2)$$

In Equations (1) and (2),  $D_x$  and  $D_y$  are distances along the  $x$  and  $y$  coordinate axes within the fragment volume,  $v_0/x_0$  is the gradient of the shear ('Couette') flow present in the fragment volume at the moment of fragmentation;  $r_2$  is the radius vector within the circular solar nebula; and  $G$  is the gravitational constant. The parameter  $a$  gives the strength of the shear flow relative to the angular velocity,  $\omega$ , of the pre-supernova star:

$$\omega = +a(v_0/x_0). \quad (3)$$

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TABLE I  
Flows in the planetary disk vs  $a$  for  $D_{ym} = D_{xm}$

| Value of $a$  | Conic locus                    | Planetary disk flows                                                                            | Planets                                  |
|---------------|--------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------|
| $a \ll 0$     | circular cylinder              | flow only in the (+) direction                                                                  | all gas giants                           |
| $a < 0$       | elliptical cylinder            | flow only in the (+) direction                                                                  | all gas giants                           |
| $a = 0$       | 2 planes                       | pure shearing flow in the (+) direction; indicates negligible stellar rotation                  | all gas giants                           |
| $0 < a < 1/2$ | 4-branched hyperbolic cylinder | opposing flows in the disk center; $r_{2m+} > r_{2m-}$ ; central material delivered to protosun | inner: small, dense<br>outer: gas giants |
| $a = 1/2$     | 4-branched hyperbolic cylinder | opposing flows throughout the disk; $r_{2m+} = r_{2m-}$ ; most material delivered to protosun   | none; possibly asteroids                 |
| $1/2 < a < 1$ | 4-branched hyperbolic cylinder | opposing flows in the disk center; $r_{2m-} > r_{2m+}$ ; central material delivered to protosun | inner: small, dense<br>outer: gas giants |
| $a = 1$       | 2 planes                       | pure shearing flow in the (-) direction; equivalent to $a = 0$ case, but in the (-) direction   | all gas giants                           |
| $a > 1$       | elliptical cylinder            | flow only in the (-) direction                                                                  | all gas giants                           |
| $a \gg 1$     | circular cylinder              | flow only in the (-) direction                                                                  | all gas giants                           |

(+) denotes counter clockwise flow; (-) clockwise flow.

As can be seen from Equation (1) the locus of points  $(D_x, D_y)$  for a constant  $r_2$  is a conic section, where the type of conic section depends on  $a$ . Both counter-clockwise (+) and clockwise (-) flows are developed within the fragment due to the combination of the shear flow and the stellar rotation.

Equation (1) maps the mass element at any point  $(D_x, D_y)$  within the fragment volume to the radius,  $r_2$ , in a disk. When a fragment shape and its orientation with respect to the shear flow is assumed, the mapping process results in a mass distribution,  $m(r_2)$  in the 'planetary disk' (solar nebula). The mass distribution  $m(r_2)$  has dimensions of mass per unit length and is related to the azimuthally symmetric surface density,  $\sigma(r_2)$ , by  $m(r_2) = 2\pi r_2 \sigma(r_2)$ . The computer program ('SOLSYS') to implement the fragment-to-disk mapping is described in Brown and Gritz (1987).

The parameter  $a$  has a profound effect on the resulting mass distribution in the disk. For certain values of  $a$  ( $0 < a < 1$ ), the counter-clockwise (+) and clockwise (-) flows clash when the material settles into the disk, and the resulting head-on collisions of the particles tend to cancel the angular momentum of the particles. The opposing flows generally clash near the disk center, thus removing the material in this region and delivering the material to the accreting central mass with little or no

angular momentum. This may help to answer the long-standing question about our solar system of how our sun came to have almost all of the system's mass but very little of its angular momentum. Because during the collision the heavy elements tend to remain in orbit while the light elements are removed (Brown, 1971b), the opposing flow hypothesis may also answer the crucial question of why our inner planets are small and dense while our outer planets are gas giants. However, limiting our attention to  $0 < a < 1$  just because this interval is appropriate to our solar system (and the Beta Pictoris disk; cf. Smith and Terrile, 1984) seems parochial: other solar systems may have formed with  $a$  outside this range. Table I summarizes the possibilities. In this table,  $D_{xm}$  and  $D_{ym}$  symbolize the maximum extent of the fragment in the  $x$  and  $y$  directions, respectively. In addition to Smith and Terrile (1984), two recent articles, present evidence for other circumsolar shells or disks (Sadakane and Nishida, 1986; Little-Marenin, 1986).

One article, pertinent to the formation of solar systems from supernovae (Clayton, 1982), was overlooked at the time the article previous to this one was written (Brown and Gritz, 1987). Clayton concludes that the presumption that a nearby supernova triggered the formation of the solar system is too constraining and should be relaxed.

## 2. Analytical Solutions

In general, owing to the complexity of the problem, the fragment-to-disk mapping must be done numerically with a computer program. However, there are certain values of  $a$  and  $\theta$  (the angle of fragment ejection measured from the pole of the pre-supernova star) for which analytical solutions for  $m(r_2)$  may be found.

The value  $a = 0$  is such a case, and the analytical solution for a fragment of right circular cylindrical shape has been presented in full in (Brown, 1970). The part of this solution apparently applicable to our solar system was presented in Brown (1971a). We may see from Equation (1) that if  $a = 0$ , then the locus of points within the fragment that map to a particular radius  $r_2$  in the disk is a pair of planes perpendicular to the  $x$ -axis at the distances  $\pm D_x$  from the origin of the coordinate system (at the center of mass of the fragment of uniform density). Hence, the flow is pure shearing flow in the counter-clockwise direction, and the analytical solution involves the intersection of a right circular cylinder tilted at an angle  $\theta$  (about the  $y$ -axis) with the planes at  $\pm D_x$ . A particularly simple solution (the derivation of which is presented in the Appendix) is obtained for  $a = 0$  and  $\theta = 90^\circ$ : i.e.,

$$\frac{r_2 m(r_2)}{M_p} = \frac{1}{4} \left( \frac{r_2}{r_{2m}} \right)^{1/4} \quad (4)$$

where  $r_{2m}$  is the radius of the disk rim (corresponding to  $D_{xm}$  in the fragment);  $M_p$  is the mass of the planetary disk; and the quantity  $r_2 m(r_2)$  expresses the mass per logarithmic interval in  $r_2$ .

Another useful result which was revealed in the analysis of the  $a = 0$  case is that the solutions are parameterized by the combination of variables  $(b/\epsilon) \tan \theta$  (Brown,

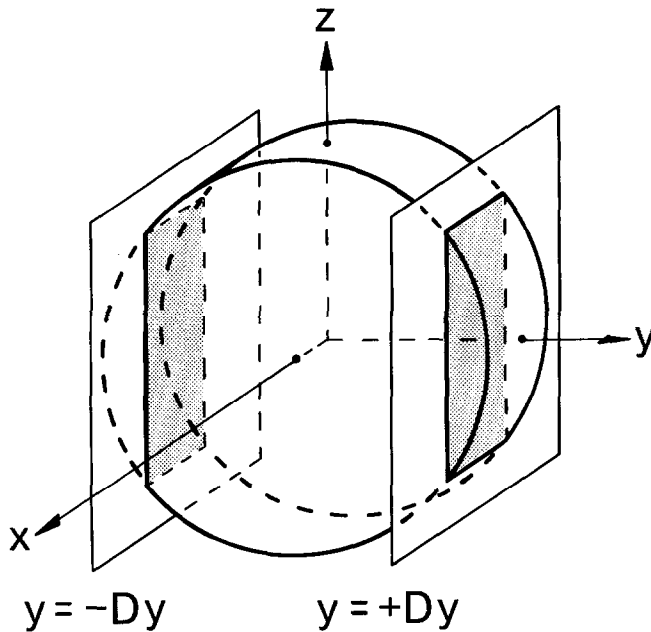


Fig. 1. The geometrical arrangement for the case  $a = 1$ . The cylinder is shown here positioned at  $\theta = 90^\circ$  for convenience only. The shaded areas of the two planes at  $y = \pm D_y$  are the locus of points of constant angular momentum.

1970) where  $2b$  is the height and  $\kappa$  is the radius of the cylindrical fragment (the ends of the cylinder are coincident with the inner and outer surfaces of the supernova shell). It was later shown numerically that the  $(b/\kappa) \tan \theta$  scaling held approximately for  $a \neq 0$  (forward flow) solutions as well (Brown and Gritz, 1987). The fact that the mass distributions are scaled with the combination  $(b/\kappa) \tan \theta$  effectively reduces the number of variables and (as will be seen in the next section) greatly facilitates the study of other solar systems.

Unfortunately, we see from Equation (3) that if we allow  $a = 0$ , it means that the pre-supernova star was not rotating, and I consider that an unlikely situation.

A more realistic case for which an analytical solution is readily available is that for  $a = 1$ . In this case, we see from Equation (1) that the locus of points which map to a particular radius,  $r_2$ , in the disk is a pair of planes perpendicular to the  $y$ -axis at distances of  $\pm D_y$  from the origin. This situation is depicted in Figure 1 for a right circular cylindrical fragment. Note that rotation through  $\theta$  (about the  $y$ -axis) does not change the area within the fragment in the two planes at  $\pm D_y$ , and thus that the mass distribution in the disk,  $m(r_2)$  is not a function of  $\theta$ . The analytical solution in this case was presented in (Brown, 1970) and is given by

$$\frac{r_2 m(r_2)}{M_p} = \frac{1}{\pi} \left( \frac{r_2}{r_{2m}} \right)^{1/4} \sqrt{1 - \left( \frac{r_2}{r_{2m}} \right)^{1/2}}. \quad (5)$$

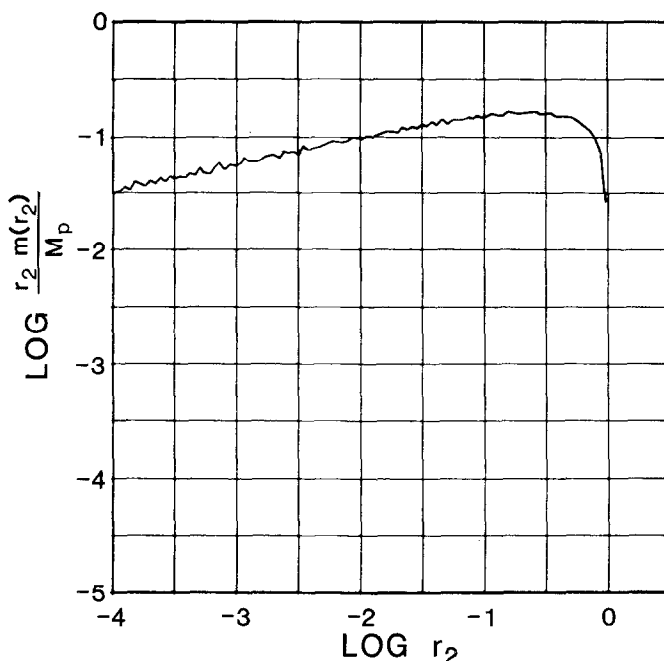


Fig. 2. The disk mass distribution for the case  $a = 1$ , (Equation (5) in the text). This solution is not a function of the dimensions ( $b$ ,  $\iota$ ) of the cylindrical fragment, or of its angle of ejection ( $\theta$ ).

The derivation of Equation (5) may be found in the Appendix. The mass distribution given by Equation (5) is shown in Figure 2.

The above analytical solutions for  $a = 0, 1$  were used to verify the mapping program described in Brown and Gritzko (1987).

Another special case for which an analytical solution is readily found is that at the pole of the pre-supernova star, or  $\theta = 0^\circ$ . Although the shearing flow at the equator ( $\theta = 90^\circ$ ) is well represented by a parallel ('Couette') flow, this approximation is not good at the pole, where there is only circular flow. This results from a combination of circular shearing flow and rotation of the pre-supernova star. The pole solution is derived in the Appendix, and is given by

$$\frac{r_2 m(r_2)}{M_p} = \frac{1}{2} \left( \frac{r_2}{r_{2m}} \right)^{1/2}. \quad (6)$$

This solution is also applicable near the centers of disks for which  $a \ll 1$  or  $a \gg 1$ , where the locus of points (for a fixed  $r_2$ ) in the fragment approaches a circular cylinder (see Equation (1)).

### 3. Other Solar Systems

It has been shown that the present model of solar system formation provides a

TABLE II  
Examples chosen for other solar systems

| $\begin{matrix} a \\ (b/\iota) \tan \theta \end{matrix}$ | -2.0    | -1.0    | 0       | +0.3    | +0.7    | +1.0    | +2.0    | +3.0    |
|----------------------------------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|
| 0 ( $\theta = 0^\circ$ )                                 | Eq. (6) | Eq. (6) | Eq. (6) | Eq. (6) | Eq. (6) | Eq. (6) | Eq. (6) | Eq. (6) |
| 0.25                                                     | 3a      | 3b      | A       | 3c      | 3d      | Eq. (5) | 3e      | 3f      |
| 0.50                                                     | 4a      | 4b      | A       | 4c      | 4d      | Eq. (5) | 4e      | 4f      |
| 0.75                                                     | 5a      | 5b      | A       | 5c      | 5d      | Eq. (5) | 5e      | 5f      |
| 1.0                                                      | 6a      | 6b      | A       | 6c      | 6d      | Eq. (5) | 6e      | 6f      |
| 1.5                                                      | 7a      | 7b      | A       | 7c      | 7d      | Eq. (5) | 7e      | 7f      |
| 2.0                                                      | 8a      | 8b      | A       | 8c      | 8d      | Eq. (5) | 8e      | 8f      |
| 3.0                                                      | 9a      | 9b      | A       | 9c      | 9d      | Eq. (5) | 9e      | 9f      |
| $\infty$ ( $\theta = 90^\circ$ )                         | 10a     | 10b     | Eq. (4) | 10c     | 10d     | Eq. (5) | 10e     | 10f     |

A = Analytical; found in (Brown, 1970, 1971a);  $\theta \geq 20^\circ$ .

reasonably good description of the mass distribution of our own solar system for certain values of the model parameters ( $(b/\iota) \tan \theta = 0.44$ ,  $0.1 \leq a(\theta) \leq 0.5$ ; Brown and Gritzo, 1987). To examine the mass distributions in the planetary disk of other solar systems, we must explore the parameter space of the model.

The limits on the angle of fragment ejection are  $0^\circ \leq \theta \leq 90^\circ$ . The limits on the aspect ratio of the cylindrical (or prismatic) fragment may be estimated to be  $0.1 \leq b/\iota \leq 2$ , where we reason that the fragment might be quite flat, but not too rod-shaped. The limits on the parameter  $a$  can only be guessed. For practical reasons I will set  $-2 \leq a \leq +3$ , but from Equation (3), if  $v_0/x_0$  is small,  $a$  might be large.

Table II presents a matrix of the cases I have chosen to investigate for the present study. In Table II may be found either the equation number of the analytical mass distribution or the Figure number (3 through 10) of the mass distribution numerically calculated with the 'SOLSYS' program (Brown and Gritzo, 1987).

In Figures 3a through 10f, the abscissa and ordinate scales are identical to those on Figure 2: they all have a grid line every one-half decade. Although all the plots look roughly similar, I emphasize that Figures 2 through 10 are log-log plots, which causes a deceptive compression of the scales (both  $\log r_2$  (abscissa) and  $\log r_2 m(r_2)$  (ordinate)).

Some specific comments on Figures 3 through 10 follow: Figure 4c is like our solar system (Brown and Gritzo, 1987) and Figures 3c and 5c are similar. In Figure 7d, the value of  $a$  is such that both the reverse flow (dashed line) and the forward flow (solid line) are in evidence. In this case the outermost planets would be revolving about their sun in the opposite direction than the inner planets. Some of the plots exhibit gaps near the disk center. This indicates that a small amount of material is revolving in the opposite direction in this region.

Figures 2 through 10 depict the possible mass distributions in the solar nebulae of other solar systems. Finding the masses of the planets that will form is a separate problem. If the annular bands that form in the disks (that will lead to the formation

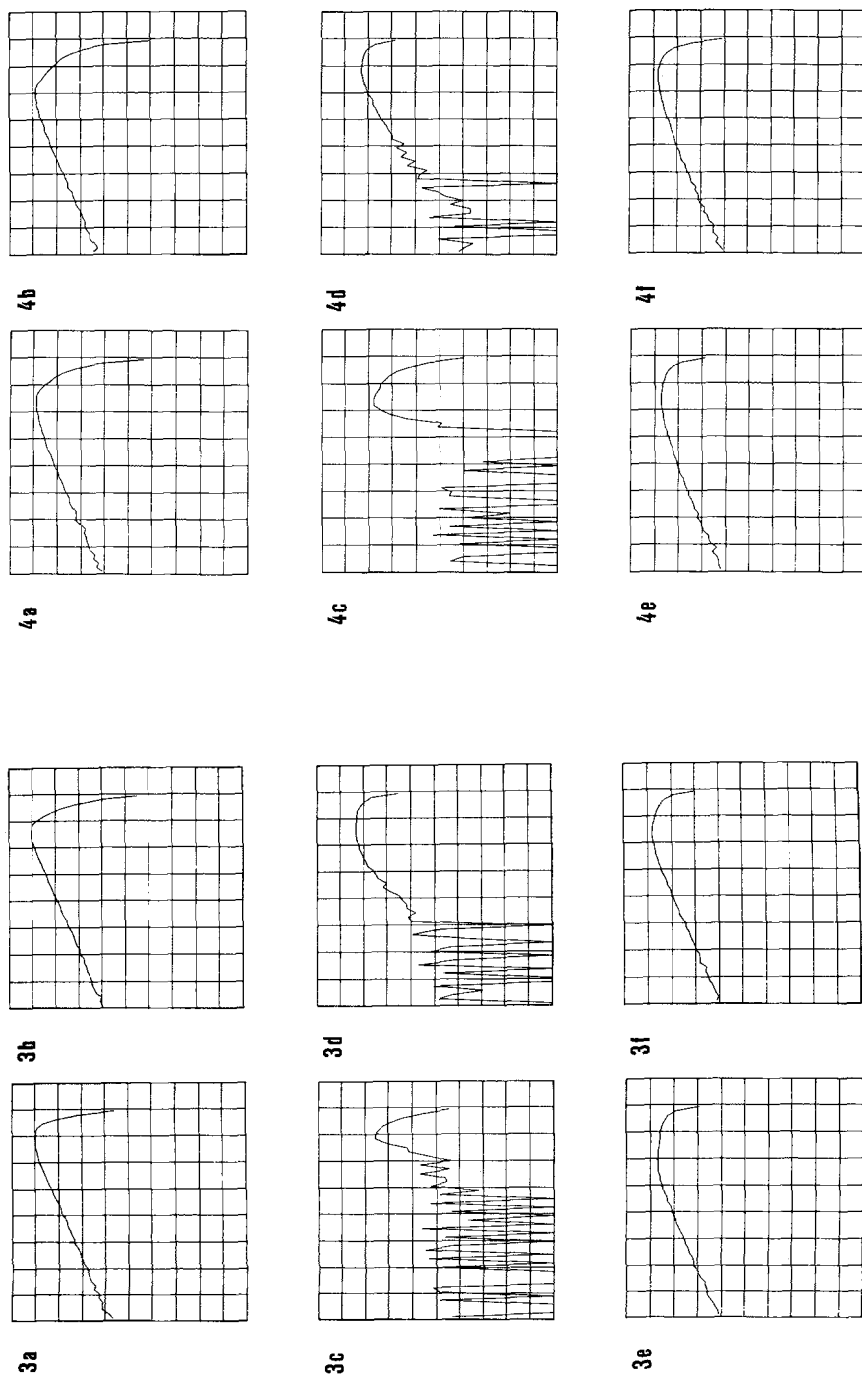
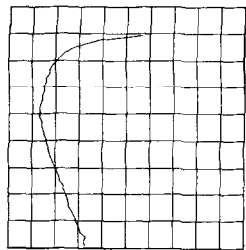
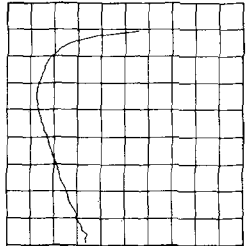


Fig. 3. The disk mass distribution for  $(b, \epsilon)$   $\tan \theta = 0.25$ . See Table II for key. The ordinate and abscissa scales of each of the six plots are identical to those in Figure 2.

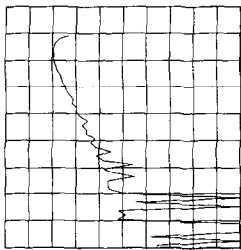
Fig. 4. The disk mass distributions for  $(b, \epsilon)$   $\tan \theta = 0.50$ . See Table II for key. The ordinate and abscissa scales of each of the six plots are identical to those of Figure 2.



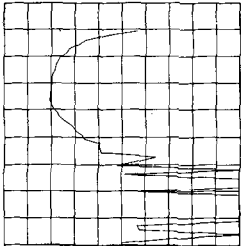
6a



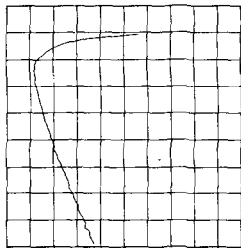
6b



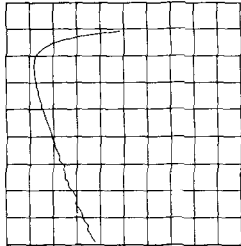
6c



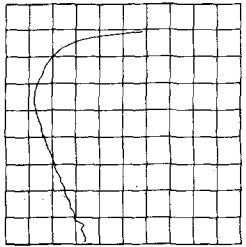
6e



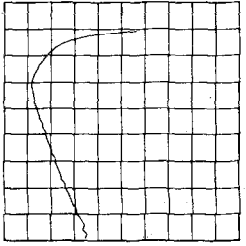
6f



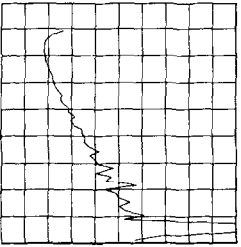
6g



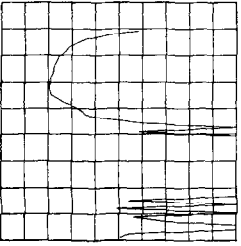
5a



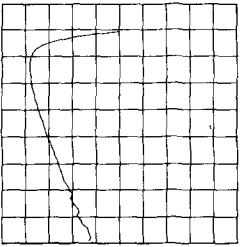
5b



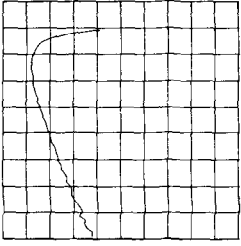
5c



5e



5f



5g

Fig. 6. The disk mass distribution for  $(b/\epsilon) \tan \theta = 1.0$ . See Table II for key. The ordinate and abscissa scales of each of the six plots are identical to those of Figure 2.

Fig. 5. The disk mass distributions for  $(b/\epsilon) \tan \theta = 0.75$ . See Table II for key. The ordinate and abscissa scales of each of the six plots are identical to those of Figure 2.



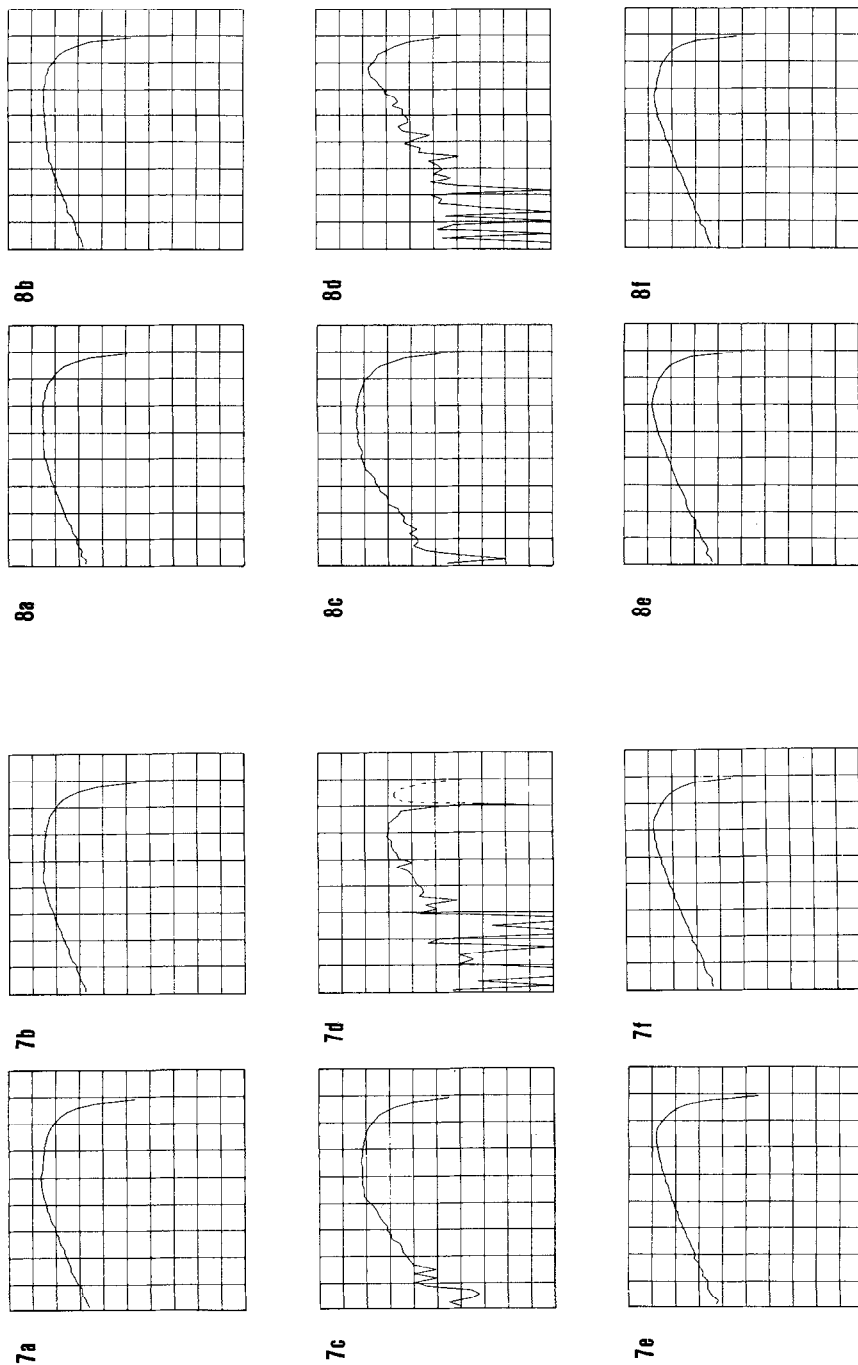


Fig. 7. The disk mass distribution for  $(b/\lambda) \tan \theta = 1.5$ . See Table II for key. The ordinate and abscissa scales of each of the six plots are identical to those of Figure 2.

Fig. 8. The disk mass distribution for  $(b/\lambda) \tan \theta = 2.0$ . See Table II for key. The ordinate and abscissa scales of each of the six plots are identical to those of Figure 2.

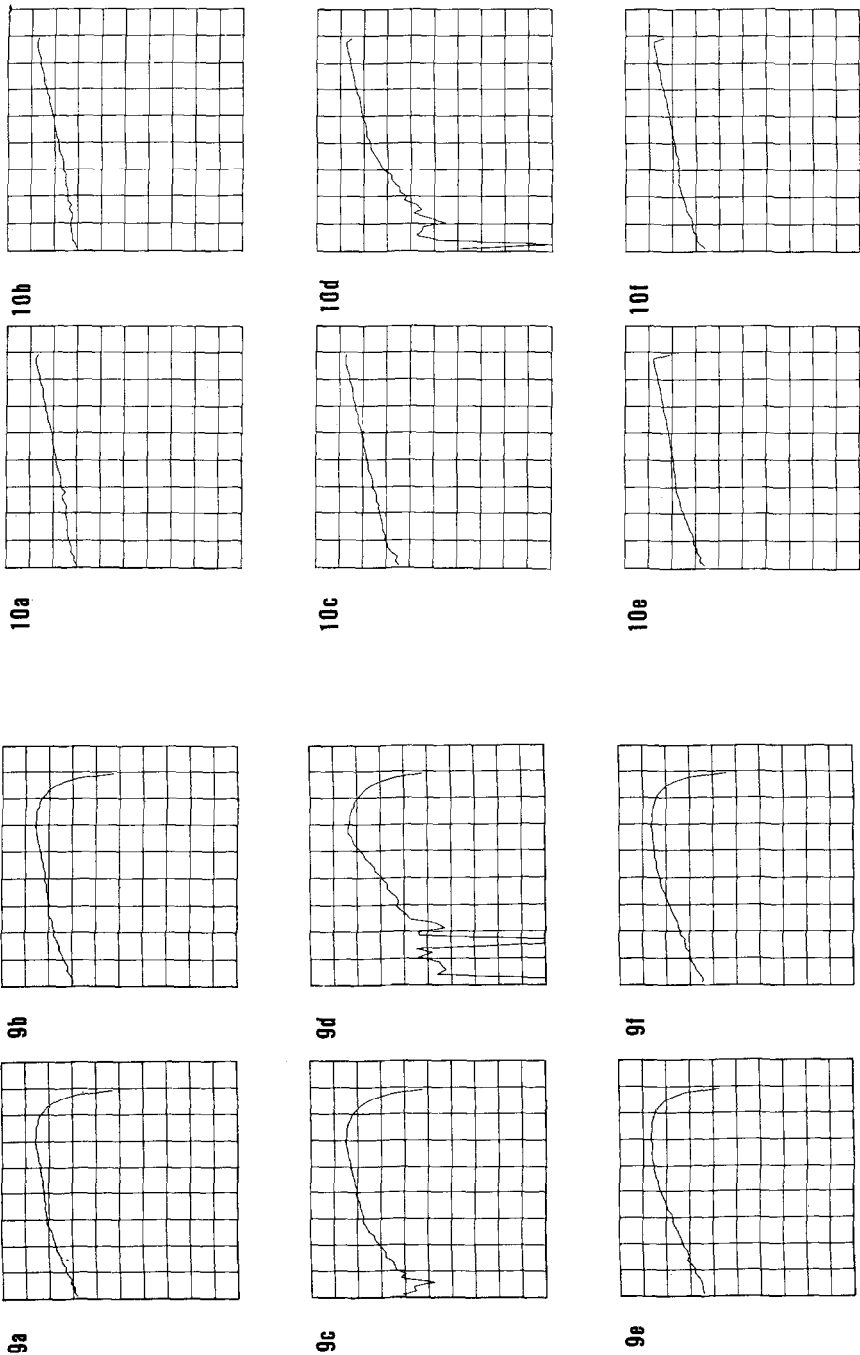


Fig. 9. The disk mass distribution for  $(b/\lambda) \tan \theta = 3.0$ . See Table II for key. The ordinate and abscissa scales of each of the six plots are identical to those of Figure 2.

Fig. 10. The disk mass distribution for  $(b/\lambda) \tan \theta = \infty$ . See Table II for key. The ordinate and abscissa scales of each of the six plots are identical to those of Figure 2.

of planets) are logarithmically spaced, then Figures 2 through 10 also indicate the masses of the planets. This is because both the ordinate,  $r_2 m(r_2)$ , and the logarithmically spaced planets indicate the mass distribution in terms of mass per logarithmic interval. The planets in our solar system are approximately logarithmically spaced, but if the planets of other solar systems are not so spaced, then Figures 2 through 10 are *not* indicative of the masses of those planets – but only indicative of the mass distributions in the solar nebulae.

In this regard, it was proposed in (Brown, 1970, 1971a) that the vertices of a prismatic fragment would give rise to maxima in the mass distribution of the solar nebula. Such maxima would predispose the disk to form planets at those radii. In the case of our solar system, it was found that  $(b/\lambda) \tan \theta = 0.43$  gave planetary spacings that were quite like that of our planets. This value of  $(b/\lambda) \tan \theta$  is also the one that leads to a mass distribution most like our solar system. However, when the calculated mass distribution was closely examined for the existence of these possible maxima, they were not in evidence – at least they must have been smaller than  $\sim 0.5\%$  (Brown and Gritz, 1986). So, apparently the present model of solar system formation does not include a mechanism that dictates the spacing of planets. Other authors (Dole, 1970; Isaacman and Sagan, 1977; Patterson, 1986) have presented promising pictures of how planets might achieve their spacing.

Another factor that has a profound influence on the morphology of a forming solar system is the rim radius, given by the larger of either

$$r_{2m+} = \frac{(v_0/x_0)^2}{G M_c} (1-a)^2 D_{xm}^4 \quad (7)$$

or

$$r_{2m-} = \frac{(v_0/x_0)^2}{G M_c} a^2 D_{ym}^4 \quad (8)$$

where  $M_c$  is the dominant central mass (Brown and Gritz, 1986).

Noting that  $\rho \propto M_p/\pi D_{xm}^2 (2D_{xm}) \propto M_p/D_{ym}^2 (2D_{ym})$ , and assuming that  $M_c \propto M_p$ , we have

$$r_{2m+} \propto \frac{1}{2\pi G \rho} \left(\frac{v_0}{x_0}\right)^2 D_{xm} (1-a)^2 = \frac{\omega^2 D_{xm}}{2\pi G \rho} \left(\frac{1}{a} - 1\right)^2, \quad (9)$$

and

$$r_{2m-} \propto \frac{1}{2\pi G \rho} \left(\frac{v_0}{x_0}\right)^2 D_{ym} a^2 = \frac{\omega^2 D_{ym}}{2\pi G \rho}. \quad (10)$$

It can be seen that the rim radius can vary over a wide range that includes zero.

### Summary and Conclusions

The mass distributions that may be found in the solar nebulae of other solar systems have been presented. The variable  $a$  – which expresses the ratio of shear flow to the rotation of the pre-supernova star – is found to be crucial to the shape of the mass distributions. Unfortunately, the limits on this parameter are the least well established of all the model variables. Because of this, the range of  $a$  may have been overestimated. Further experimental and theoretical work may improve the constraints on the parameter  $a$ .

### Appendix: Special Cases

#### 1. The Pole Solution ( $\theta = 0^\circ$ , any $a$ )

At the pole of the supernova shell ( $\theta = 0^\circ$ ) the combined motion of shearing and stellar rotation is all circular. The appropriate solution is presented below.

Based on the mapping of the cylindrical area  $A(D)$  at radius  $D$  in the fragment (which contains all points having the particular angular momentum per unit mass necessary to wind up at radius  $r_2$  in the disk) we may write

$$m(r_2)dr_2 = \rho A(D)dD, \quad (\text{A1})$$

that expresses conservation of mass. Here  $m(r_2)$  is the disk (solar nebula) mass distribution in dimensions of mass per unit length along the radial vector  $r_2$ , and  $\rho$  is the fragment density (taken to be uniform). Setting  $\rho = M_p/\pi D_m^2 2b$  (where  $D_m$  is the radius of the cylindrical fragment;  $2b$  is the height of the cylinder; and  $M_p$  is the total mass of the disk), setting the cylindrical surface area,  $A(D) = 2\pi D 2b$ , and rearranging Equation (A1) we have

$$\frac{m(r_2)}{M_p/r_{2m}} = 2 \left( \frac{D}{D_m} \right) \frac{dD/D_m}{dr_2/r_{2m}}, \quad (\text{A2})$$

where  $r_{2m}$  is the radius of the rim of the circular disk corresponding to  $D_m$  in the fragment.

Needed now is a map from the radius  $D/D_m$  in the fragment to the radius  $r_2/r_m$  in the circular disk. To obtain this, we will conserve angular momentum per unit mass in the fragment-to-disk transition. We assume that the fragment at the pole is in rigid rotation so that the tangential velocity,  $v(D)$ , in the fragment is given by

$$v(D) = \omega_0 D, \quad (\text{A3})$$

where  $\omega_0$  is the angular velocity (a constant). The angular momentum per unit mass is given by  $Dv(D)$  so that from Equation (A3)

$$Dv(D) = \omega_0 D^2 = r_2 v(r_2), \quad (\text{A4})$$

where the quantity on the right-hand side is the angular momentum per unit mass

in the disk. The tangential velocity  $v(r_2)$  in the disk is dictated by requiring orbital balance, which is given by

$$v(r_2) = \sqrt{\frac{G M_c}{r_2}}, \quad (\text{A5})$$

where  $M_c$  is the dominant central mass in the disk. Combining Equations (A4) and (A5) we have

$$\omega_0 D^2 = \sqrt{G M_c} r_2^{1/2}. \quad (\text{A6})$$

To write Equation (A6) for the disk rim, we replace  $D$  with  $D_m$  and  $r_2$  with  $r_{2m}$ . Then, dividing Equation (A6) by this rim equation, we obtain the fragment-to-disk map of

$$\frac{D}{D_m} = \left( \frac{r_2}{r_{2m}} \right)^{1/4}. \quad (\text{A7})$$

Equation (A7) furnishes the quantities needed in Equation (A2), and inserting these quantities into the latter equation, we have

$$\frac{m(r_2)}{M_p/r_{2m}} = {}^{1/2} \left( \frac{r_2}{r_{2m}} \right)^{-1/2}, \quad (\text{A8})$$

or, in terms of mass per logarithmic interval in  $r_2$ ,

$$\frac{r_2 m(r_2)}{M_p} = {}^{1/2} \left( \frac{r_2}{r_{2m}} \right)^{1/2}. \quad (\text{A9})$$

This mass distribution for the pole of the supernova shell is presented in the text of this article as Equation (6).

## 2. The Reverse Shear Solution ( $a = 1$ , any $\theta$ )

In this case,  $D_y$  is a distance along the  $y$ -axis, and the locus of points of constant angular momentum is a pair of planes perpendicular to the  $y$ -axis at  $\pm D_y$ . Reference to Equation (1) and Figure 1 may be helpful. Here  $A(D_y)$  involves the length of a chord in the cylinder perpendicular to  $D_y$ , such that  $A(D_y) = 2(D_{ym}^2 - D_y^2)^{1/2} 2b$ . Substituting  $A(D_y)$  and the value of  $\rho$  given previously into Equation (1) we obtain

$$\frac{m(r_2)}{M_p/r_{2m}} = \frac{4}{\pi} \sqrt{1 - \left( \frac{D_y}{D_{ym}} \right)^2} \frac{dD_y/D_{ym}}{dr_2/r_{2m}}. \quad (\text{A10})$$

Again, we need a map from  $\pm D_y$  in the cylindrical fragment to radius  $r_2$  in the disk having the same angular momentum per unit mass. The shearing ('Couette') flow anti-parallel to the  $x$ -axis is given by

$$v_x(D_y) = -\frac{v_0}{y_0} D_y, \quad (\text{A11})$$

where  $v_0/y_0$  is the gradient of the shear flow. Conservation of angular momentum per unit mass gives

$$D_y v_x(D_y) = -\frac{v_0}{y_0} D_y^2 = r_2 v(r_2), \quad (\text{A12})$$

where  $v(r_2)$  is dictated by orbital balance in the disk, Equation (A5).

We proceed as before: Equation (A5) is substituted in Equation (A12) and we obtain an equation essentially the same as Equation (A6), but with  $v_0/y_0$  appearing in place of  $\omega_0$ , and  $D_y$  appearing in place of  $D$ . Dividing Equation (A12) by the equation evaluated at the rim, we again obtain Equation (A7). Substituting Equation (A7) into Equation (A10), we obtain the mass distribution in the disk

$$\frac{m(r_2)}{M_p/r_{2m}} = \frac{1}{\pi} \left( \frac{r_2}{r_{2m}} \right)^{-3/4} \sqrt{1 - \left( \frac{r_2}{r_{2m}} \right)^{1/2}}. \quad (\text{A13})$$

In terms of mass per logarithmic interval in  $r_2$ ,

$$\frac{r_2 m(r_2)}{M_p} = \frac{1}{\pi} \left( \frac{r_2}{r_{2m}} \right)^{1/4} \sqrt{1 - \left( \frac{r_2}{r_{2m}} \right)^{1/2}}. \quad (\text{A14})$$

This mass distribution for the case  $a = 1$  is presented in the text as Equation (5).

### 3. The Equator Solution ( $a = 0$ , $\theta = 90^\circ$ )

For a cylindrical fragment ejected from the supernova at  $\theta = 90^\circ$ , for  $a = 0$  the area  $A(D_x)$  is a constant (the area of the circular face of the cylinder) and we find from Equation (A1) that

$$\frac{m(r_2)}{M_p/r_{2m}} = \frac{dD_x/D_{xm}}{dr_2/r_{2m}}. \quad (\text{A15})$$

Using Equation (A7) we obtain

$$\frac{m(r_2)}{M_p/r_{2m}} = {}^{1/4} \left( \frac{r_2}{r_{2m}} \right)^{-3/4}. \quad (\text{A16})$$

In terms of mass per unit logarithmic interval, the mass distribution is

$$\frac{r_2 m(r_2)}{M_p} = \frac{1}{4} \left( \frac{r_2}{r_{2m}} \right)^{1/4}. \quad (\text{A17})$$

This is presented as Equation (4) in the text.

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